

Supplementary material (Online Resource 1) for “Detectability-driven design of condition monitoring for three types of wind-turbine generators under identical winding faults”

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Abstract

This supplement contains the feature definitions (Section S1), the computation details of the demonstration (Section S2), the supplementary tables S1–S12 and the supplementary figures S1–S2 referred to in the main text. Every table and figure is produced by the released code from the public datasets.

Keywords: supplementary material

S1 Feature definitions

Let x_a, x_b, x_c be the sampled phase quantities of a window of N samples at sampling frequency f_s , w the Hann window and f_e the electrical frequency of the window. The complex fundamental amplitude of a signal x at frequency f is

$$X(f) = \frac{2}{\sum_n w_n} \sum_{n=0}^{N-1} x_n w_n e^{-j2\pi f n / f_s}. \quad (1)$$

The electrical frequency is estimated from the slope of the unwrapped phase of the current space vector $\frac{2}{3}(i_a + \mathbf{a} i_b + \mathbf{a}^2 i_c)$, $\mathbf{a} = e^{j2\pi/3}$, whose sign gives the phase rotation; outside 0.5–1.5 times the nominal value the nominal value is used. Sequence magnitudes are $|X_+| = |X_a + \mathbf{a} X_b + \mathbf{a}^2 X_c|/3$ and $|X_-| = |X_a + \mathbf{a}^2 X_b + \mathbf{a} X_c|/3$ for an

abc rotation (interchanged for acb), evaluated at $f = f_e$ for currents (I_1, I_2), terminal voltages (V_1, V_2) and mean-removed duty cycles (D_1, D_2); $|X(f)|$ is a fundamental amplitude (peak) and is divided by $\sqrt{2}$ wherever it is compared with an RMS value. The RMS unbalance is $\max_k |I_{\text{rms},k} - \bar{I}_{\text{rms}}| / \bar{I}_{\text{rms}}$ over the three phases. Harmonic ratios are $|I_a(kf_e)| / |I_a(f_e)|$ for $k = 3, 5, 7$, and the THD is $\sqrt{\sum_{k=2}^{20} |I_a(kf_e)|^2} / |I_a(f_e)|$. For the synchronous-frame currents i_d, i_q and the torques the features are the standard deviation of the mean-removed signal and its amplitudes $|Y(f_e)|$ and $|Y(2f_e)|$; for the PI outputs the standard deviation and $|Y(2f_e)|$; for the measured and commanded synchronous-frame voltages $|Y(2f_e)|$ only; for the field current the standard deviation and $|Y(2f_e)|$. Speed and DC-link features are standard deviations. Windows are n_c electrical cycles long with a hop of one cycle; a feature is undefined for a trial whose faulted interval holds no complete window. On the WFSG bench the measured converter-side phase voltages serve as terminal voltages, the synchronous-frame voltages are obtained from them with the encoder angle, the commanded voltages are the controller's, and the electrical torque is computed from power and speed.

S2 Computation details

Intervals.

PMSG and SCIG: reference interval 0.30–0.98 s after the start of each record, split at 0.62 s into R_1 (0.32 s) and R_2 (0.36 s); contactor commanded at 1.000 s; faulted interval from 30 ms after the command to 10 ms before its release; the relay picks up 85–90 ms after the command, so the faulted interval contains about 55 ms of healthy time (at most 15 % of its length); recovery windows start 0.30 s after the release. WFSG: fault interval located from the measured fault current as the first contiguous interval in which its 10 ms RMS exceeds $\max(0.3 \text{ A}, 6\sigma)$, σ being the standard deviation of that RMS over 0.05–0.40 s, ending at the first fall below the threshold after 50 ms; reference interval from 0.05 s after the start of the record to 10 ms before the onset, split at 0.30 s (R_1 0.25 s, R_2 about 0.24 s); recovery windows start 0.15 s after the end of the fault interval. At 23.9–63.7 Hz a WFSG faulted interval holds between zero and eleven three-cycle windows; four trials hold none and are dropped; 81 of 438 hold fewer than three (all 50 trials at 1671 rpm, 26 of 206 at 1800 rpm, 4 at 716 and 1 at 955 rpm).

Null and screen.

The null is pooled over all 225 records of each of the PMSG and SCIG and over the 438 inter-turn and inter-winding records of the WFSG. The reliability screen flags a feature when its count of false alarms on the nine healthy trials is incompatible with the nominal rate $1 - \alpha$ at the 5 % level (one-sided binomial test: three or more at $\alpha = 0.95$, two or more at $\alpha = 0.99$, four or more at $\alpha = 0.90$), and excludes features whose null quantile exceeds one.

Bootstrap.

2000 resamples, seed 1. Median SDR and detectable share of a class: cluster bootstrap over the twelve tap pairs, the same resampled pairs for both machines; Wilson

95 % intervals for shares. Minimum detectable extent: operating-point trials resampled within each extent level; the limit-of-detection rule applied to each resample. The null quantile is held at its full-sample value. Decision table: 1000 resamples per suite, alternative, class and α (seed 7); a suite *meets* e_{req} when the resample probability of $\text{MDE} \leq e_{\text{req}}$ is at least 0.95 and *fails* when it is at most 0.05.

Transfer.

Windows: reference, faulted and recovery windows of all trials; positives: faulted windows of fault trials. Models: scikit-learn histogram gradient boosting (max depth 3, 200 iterations, learning rate 0.08, random state 0) and logistic regression with standardisation ($C = 1$, 3000 iterations). Cross-validation: five-fold GroupKFold with one group per tap pair and machine, healthy trials grouped by speed. Referencing rules: physical units; z -score on the target machine’s healthy trials; absolute z -score against the mean and standard deviation of the first half of the trial’s own reference interval (windows starting before 0.62 s for the PMSG/SCIG, before 0.30 s for the WFSG). Metrics: area under the receiver operating characteristic curve (AUC) and true-positive rate (TPR) at 1 % false-positive rate (FPR) on all windows and on the windows outside the normalising set; the false-alarm rate of the all-negatives 1 % threshold on the fault-time windows of the healthy trials.

S3 Supplementary tables

Table S1 Observation suites for the three alternatives, three-cycle windows, common feature set, $\alpha = 0.95$

Suite	Machine	Feature	MDE	DS	DS nested	Oracle DS	Oracle FA
<i>Inter-turn faults</i>							
drive-internal	PMSG	D_2/D_1	7.4	65	65	82	44
drive-internal	SCIG	$I_d 2f_e$	7.4	78	78	93	78
drive-internal	WFSG	$I_q 2f_e$	11.6 ^a	91	87	96	–
3 CT	PMSG	I_2/I_1	7.4	57	57	78	11
3 CT	SCIG	I_2/I_1	7.4	70	70	86	56
3 CT	WFSG	$I_a h3$	11.6 ^a	98	86	94	–
3 CT + 3 VT	PMSG	V_2/V_1	3.8	73	73	85	33
3 CT + 3 VT	SCIG	I_2/I_1	7.4	70	70	89	56
3 CT + 3 VT	WFSG	V_2/V_1	11.6 ^a	98	98	94	–
3 CT + 3 VT + drive	PMSG	V_2/V_1	3.8	73	68	94	67
3 CT + 3 VT + drive	SCIG	$I_d 2f_e$	7.4	78	78	97	89
3 CT + 3 VT + drive	WFSG	V_2/V_1	11.6 ^a	98	98	98	–
excitation (field current)	WFSG	$I_f 2f_e$	11.6 ^a	86	86	74	–
<i>Inter-winding faults</i>							
drive-internal	PMSG	D_2/D_1	14.3	91	91	98	44
drive-internal	SCIG	$I_d 2f_e$	2.3	100	100	100	78
drive-internal	WFSG	$I_q 2f_e$	2.3	100	100	100	–
3 CT	PMSG	I_2/I_1	9.7	83	83	99	11
3 CT	SCIG	I_2/I_1	2.3	98	98	99	56
3 CT	WFSG	I_2/I_1	2.3	100	100	100	–
3 CT + 3 VT	PMSG	V_2/V_1	9.7	92	92	99	33
3 CT + 3 VT	SCIG	I_2/I_1	2.3	98	98	100	56
3 CT + 3 VT	WFSG	V_2/V_1	2.3	100	100	100	–
3 CT + 3 VT + drive	PMSG	V_2/V_1	9.7	92	92	100	67
3 CT + 3 VT + drive	SCIG	$I_d 2f_e$	2.3	100	100	100	89
3 CT + 3 VT + drive	WFSG	$I_q 2f_e$	2.3	100	100	100	–
excitation (field current)	WFSG	$I_f 2f_e$	2.3	100	100	100	–

Fixed feature: highest median SDR among the suite’s screened features. MDE (%) at the fault impedance matching the PMSG/SCIG protocol (WFSG: 198 trials, four tap pairs per class). DS: detectable share (%) of the fixed feature; nested: fixed feature chosen on the other tap pairs; oracle: per-trial best feature, with its false-alarm rate on the healthy trials (FA, %; not available for the WFSG). ‘–’: largest tested extent not detectable.

^aThe smallest inter-turn extent tested on the WFSG at that impedance is 11.6%, so the value is an upper bound.

Table S2 Five features with the highest median SDR per alternative and fault class, three-cycle windows; detectable share (%) in parentheses

Machine	Class	Features: median SDR (DS)
PMSG	inter-turn	V_2/V_1 2.5 (73); D_2/D_1 2.4 (65); I_2/I_1 1.4 (57); $V_q 2f_e$ 1.2 (55); $V_q^{\text{cmd}} 2f_e$ 1.1 (55)
PMSG	inter-winding	V_2/V_1 51.5 (92); D_2/D_1 28.7 (91); $V_q 2f_e$ 19.8 (89); I_2/I_1 11.5 (83); $V_d 2f_e$ 5.6 (71)
SCIG	inter-turn	$I_d 2f_e$ 6.5 (78); $V_d^{\text{cmd}} 2f_e$ 5.5 (73); D_2/D_1 5.5 (78); I_2/I_1 4.0 (70); $V_q^{\text{cmd}} 2f_e$ 3.6 (66)
SCIG	inter-winding	$I_d 2f_e$ 73.0 (100); $V_d^{\text{cmd}} 2f_e$ 57.3 (100); D_2/D_1 50.7 (100); I_2/I_1 49.5 (98); V_2/V_1 40.7 (100)
WFSG	inter-turn	V_2/V_1 9.4 (80); $I_q 2f_e$ 6.8 (81); $V_q^{\text{cmd}} 2f_e$ 6.6 (80); RMS unbal. 5.7 (80); $V_q 2f_e$ 4.7 (81)
WFSG	inter-winding	$I_q 2f_e$ 30.9 (100); $V_q^{\text{cmd}} 2f_e$ 29.7 (100); D_2/D_1 25.7 (100); I_q std 23.1 (95); I_a h3 21.3 (100)

WFSG at all its fault impedances, trials with at least one full fault window.

^aFails the reliability screen for that machine (three or more false alarms on the nine healthy trials, or null quantile above one) and is not used in the suite tables.

Table S3 Best-feature median SDR / detectable share (%) per speed–torque cell, PMSG and SCIG

Machine	Speed (rpm)	5.2 N m	6.4 N m	8.0 N m
PMSG	1200	7.9 / 83	9.0 / 83	5.3 / 79
PMSG	1500	12.8 / 96	11.3 / 96	9.6 / 92
PMSG	1800	20.5 / 92	18.2 / 96	15.5 / 92
SCIG	1200	18.1 / 92	16.6 / 92	16.1 / 96
SCIG	1500	26.4 / 92	23.5 / 92	23.8 / 92
SCIG	1800	51.0 / 100	47.2 / 100	44.0 / 96

Best of V_2/V_1 , $V_q 2f_e$, I_2/I_1 and $PI_d 2f_e$; 24 fault trials per cell.

Table S4 Sensitivity of MDE/DS to the window length (3, 5, 8 cycles at $\alpha = 0.95$) and to the null quantile ($\alpha = 0.90$ and 0.99 at five cycles)

Class	Feature	Machine	3 c	5 c	8 c	$\alpha 0.90$	$\alpha 0.99$
IT	V_2/V_1	PMSG	3.8/73	3.8/76	3.8/73	3.8/80	3.8/69
IT	V_2/V_1	SCIG	7.4/69	7.4/73	7.4/75	7.4/76	7.4/65
IT	I_2/I_1	PMSG	7.4/57	7.4/56	7.4/58	7.4/65	7.4/52
IT	I_2/I_1	SCIG	7.4/70	7.4/75	7.4/75	7.4/79	7.4/68
IT	$I_d 2f_e$	PMSG	12.1/29	12.1/30	12.1/28	12.1/33	12.1/24
IT	$I_d 2f_e$	SCIG	7.4/78	7.4/79	7.4/83	7.4/84	7.4/72
IT	$PI_d 2f_e$	PMSG	12.1/30	12.1/31	12.1/28	12.1/33	12.1/24
IT	$PI_d 2f_e$	SCIG	7.4/78	7.4/78	7.4/83	7.4/84	7.4/72
IT	$V_q 2f_e$	PMSG	7.4/55	7.4/57	7.4/60	7.4/65	7.7/50
IT	$V_q 2f_e$	SCIG	7.4/61	7.4/60	7.4/62	7.4/64	7.4/59
IW	V_2/V_1	PMSG	9.7/92	2.3/94	2.3/94	2.3/94	9.7/91
IW	V_2/V_1	SCIG	2.3/100	2.3/100	2.3/100	2.3/100	2.3/100
IW	I_2/I_1	PMSG	9.7/83	9.7/81	9.7/82	3.2/87	9.7/79
IW	I_2/I_1	SCIG	2.3/98	2.3/100	2.3/100	2.3/100	2.3/97
IW	$I_d 2f_e$	PMSG	21.7/71	21.7/72	21.7/69	21.7/74	21.7/69
IW	$I_d 2f_e$	SCIG	2.3/100	2.3/100	2.3/100	2.3/100	2.3/100
IW	$PI_d 2f_e$	PMSG	21.7/71	21.7/72	21.7/70	21.7/74	21.7/69
IW	$PI_d 2f_e$	SCIG	2.3/100	2.3/100	2.3/100	2.3/100	2.3/100
IW	$V_q 2f_e$	PMSG	9.7/89	9.7/90	9.7/91	9.7/91	9.7/86
IW	$V_q 2f_e$	SCIG	9.7/88	9.7/92	9.7/88	9.7/95	9.7/84

Cells: MDE (%; limit-of-detection convention; ‘-’: largest extent not detectable) / DS (%). IT: inter-turn; IW: inter-winding.

^aThe feature fails the reliability screen at that setting.

Table S5 Features failing the reliability screen at $\alpha = 0.95$ per window length and machine (false alarms on the nine healthy trials in parentheses)

Window (cycles)	Machine	Excluded features (false alarms)
3	SCIG	D_2/D_1 (3)
5	PMSG	$I_d 1f_e$ (3), speed std (4)
5	SCIG	$I_a h7$ (3), D_2/D_1 (4)
8	PMSG	PI_d std (3), speed std (4)
8	SCIG	$I_q 2f_e$ (4), D_2/D_1 (5), V_{dc} std (4)

Table S6 Transfer of a GBDT detector restricted to each suite’s features (AUC), PMSG and SCIG, five-cycle windows

Suite	z on target’s healthy trials				$ z $ vs own reference			
	P→P	S→S	P→S	S→P	P→P	S→S	P→S	S→P
drive-internal	0.87	0.87	0.73	0.74	0.89	0.95	0.92	0.86
3 CT	0.85	0.87	0.74	0.62	0.86	0.90	0.91	0.82
3 CT + 3 VT	0.84	0.89	0.86	0.70	0.89	0.90	0.92	0.89
3 CT + 3 VT + drive	0.88	0.90	0.81	0.75	0.92	0.95	0.92	0.86
+ torque transducer	0.88	0.90	0.82	0.75	0.92	0.95	0.92	0.86

Left block: features z -scored on the target’s healthy trials; right block: $|z|$ against each trial’s own reference. P: PMSG; S: SCIG; P→P and S→S: five-fold cross-validation grouped by tap pair.

Table S7 Healthy baselines of the three asymmetry ratios and the incremental negative-sequence response under fault, three-cycle windows

Machine	Class	I_2/I_1	V_2/V_1	D_2/D_1	ΔI_2	ΔV_2	$\Delta V_2/\Delta I_2$
PMSG	inter-turn	0.0055	0.0009	0.0014	-0.000	-0.00	7.9 [-2.1, 16.5]
PMSG	inter-winding	0.0051	0.0009	0.0013	0.020	0.46	15.4 [10.9, 18.0]
SCIG	inter-turn	0.0079	0.0020	0.0036	0.001	0.01	6.7 [4.0, 7.9]
SCIG	inter-winding	0.0077	0.0020	0.0035	0.059	0.40	7.0 [6.6, 7.5]
WFSG	inter-turn	0.0148	0.0073	0.0039	0.003	-0.07	17.3 [11.8, 64.2]
WFSG	inter-winding	0.0148	0.0073	0.0038	0.027	0.30	10.6 [9.7, 12.8]

Baselines: median over the reference intervals of the fault trials. ΔI_2 (A, RMS) and ΔV_2 (V, RMS): median change of the negative-sequence current and terminal voltage under fault; $\Delta V_2/\Delta I_2$ in Ω , interquartile range in brackets. WFSG at the matching fault impedance.

Table S8 Within-trial versus commissioning-baseline null, five-cycle windows, $\alpha = 0.95$

Class	Feature	M.	q_{95}		Median SDR		DS	
			within	baseline	within	baseline	within	baseline
IT	V_2/V_1	P	0.086	0.315	2.99	1.05	76	50
IT	I_2/I_1	P	0.162	0.268	1.47	0.978	56	48
IT	$I_d 2f_e$	P	0.563	0.876	0.516	0.441	30	20
IT	$PI_d 2f_e$	P	0.557	0.870	0.518	0.441	31	20
IT	$V_q 2f_e$	P	0.165	0.277	1.35	0.872	57	44
IT	D_2/D_1	P	0.090	0.211	2.78	1.11	64	52
IW	V_2/V_1	P	0.086	0.315	61.9	12.5	94	83
IW	I_2/I_1	P	0.162	0.268	11.3	6.34	81	75
IW	$I_d 2f_e$	P	0.563	0.876	4.14	2.4	72	67
IW	$PI_d 2f_e$	P	0.557	0.870	4.21	2.4	72	67
IW	$V_q 2f_e$	P	0.165	0.277	22.4	10.7	90	81
IW	D_2/D_1	P	0.090	0.211	32.6	11.9	90	85
IT	V_2/V_1	S	0.033	0.111	4.17	1.45	73	59
IT	I_2/I_1	S	0.034	0.130	4.52	1.15	75	54
IT	$I_d 2f_e$	S	0.016	0.078	6.68	1.29	79	53
IT	$PI_d 2f_e$	S	0.016	0.078	6.69	1.29	78	53
IT	$V_q 2f_e$	S	0.060	0.131	2.58	1.54	60	58
IT	D_2/D_1	S	0.010	0.068	11.1	1.57	81	52
IW	V_2/V_1	S	0.033	0.111	48.2	16.1	100	94
IW	I_2/I_1	S	0.034	0.130	55.6	12.9	100	86
IW	$I_d 2f_e$	S	0.016	0.078	74	16	100	88
IW	$PI_d 2f_e$	S	0.016	0.078	74	16	100	88
IW	$V_q 2f_e$	S	0.060	0.131	24.8	11.2	92	86
IW	D_2/D_1	S	0.010	0.068	84.9	12.2	100	82

Within: reference immediately before the fault; baseline: the healthy trial at the same operating point, recorded at another time. DS: detectable share (%). IT: inter-turn; IW: inter-winding; P: PMSG; S: SCIG.

Table S9 Tap pairs and fault extents e , identical for the three alternatives

Inter-turn			Inter-winding		
Taps	Phase	e (%)	Taps	Phase	e (%)
D11–D12	A	2.7	D04–D06	A	2.3
D09–D10	A	2.8	D16–D18	A	3.2
D23–D24	A	2.8	D04–D07	A	9.7
D21–D22	A	3.8	D16–D19	A	10.6
D06–D07	A	7.4	D01–D06	A	14.3
D14–D15	B	7.4	D13–D18	A	14.8
D18–D19	A	7.4	D01–D07	A	21.7
D02–D03	B	7.7	D13–D19	A	22.2
D17–D20	C	11.6	D12–D21	A	33.7
D13–D16	A	11.6	D11–D21	A	36.4
D01–D04	A	12.0	D12–D22	A	37.5
D05–D08	C	12.1	D11–D22	A	40.2

Table S10 Null quantile of the SDR: 95th percentile of a feature’s absolute relative change between adjacent healthy intervals, pooled per machine

Feature	PMSG (5 c)	SCIG (5 c)	PMSG (3 c)	SCIG (3 c)	WFSG (3 c)
I_2/I_1	0.162	0.034	0.160	0.036	0.082
V_2/V_1	0.086	0.033	0.095	0.037	0.037
$I_d 2f_e$	0.563	0.016	0.444	0.015	0.168
$I_q 2f_e$	0.184	0.060	0.200	0.061	0.087
$V_q 2f_e$	0.165	0.060	0.176	0.061	0.099
$V_d^{\text{cmd}} 2f_e$	0.527	0.019	0.429	0.020	0.168
D_2/D_1	0.090	0.010	0.093	0.017	0.067
$I_a \text{ h3}$	0.216	0.035	0.162	0.039	0.574
$T_e 2f_e$	0.649	0.625	0.551	0.453	0.588
speed std	0.431	0.521	0.459	0.457	1.295
$I_f 2f_e$	–	–	–	–	0.339

$N = 225$ trials for the PMSG and the SCIG; $N = 438$ for the WFSG (all its inter-turn and inter-winding trials at every fault impedance). 5 c, 3 c: five- and three-cycle windows.

Table S11 Transfer between the three alternatives with a self-referenced GBDT detector, three-cycle windows, common features

Trained on	Tested on	AUC all	AUC o.o.r.	TPR o.o.r.	FA healthy (%)
PMSG	PMSG	0.86	0.85	0.62	3
SCIG	SCIG	0.92	0.91	0.76	1
WFSG	WFSG	0.97	0.97	0.93	–
PMSG	SCIG	0.88	0.88	0.60	0
PMSG	WFSG	0.97	0.98	0.89	–
SCIG	PMSG	0.81	0.80	0.35	0
SCIG	WFSG	0.95	0.96	0.86	–
WFSG	PMSG	0.86	0.84	0.52	1
WFSG	SCIG	0.91	0.89	0.61	0
SCIG+WFSG	PMSG	0.84	0.83	0.41	0
PMSG+WFSG	SCIG	0.89	0.88	0.61	0
PMSG+SCIG	WFSG	0.97	0.97	0.88	–

Features: $|z|$ against the first half of each trial’s own reference interval; fixed seed. AUC on all windows and on the out-of-reference (o.o.r.) windows; TPR: true-positive rate at 1 % false-positive rate (FPR) on the o.o.r. windows; FA: share of the healthy trials’ fault-time windows (contactor operation without a fault) above the threshold giving 1 % FPR on all negatives. Same-machine rows: five-fold cross-validation grouped by tap pair. The WFSG has no healthy trials, so its negatives are reference and recovery windows only and its rows are optimistic.

Table S12 Signature location, PMSG and SCIG, five-cycle windows: share of fault trials (%) in which each measurement group’s best screened feature has the highest SDR

Machine	Class	Stator currents	Terminal voltages	Controller/dq	Mechanical
PMSG	inter-turn	8 (I_2/I_1)	67 (V_2/V_1)	17 (D_2/D_1)	8 (V_{dc} std)
PMSG	inter-winding	6 (I_2/I_1)	88 (V_2/V_1)	4 (D_2/D_1)	3 (V_{dc} std)
SCIG	inter-turn	28 (I_2/I_1)	13 (V_2/V_1)	56 ($PI_d 2f_e$)	4 (V_{dc} std)
SCIG	inter-winding	33 (I_2/I_1)	0 (V_2/V_1)	67 ($PI_d 2f_e$)	0 (speed std)

Representative feature in parentheses.

S4 Supplementary figures

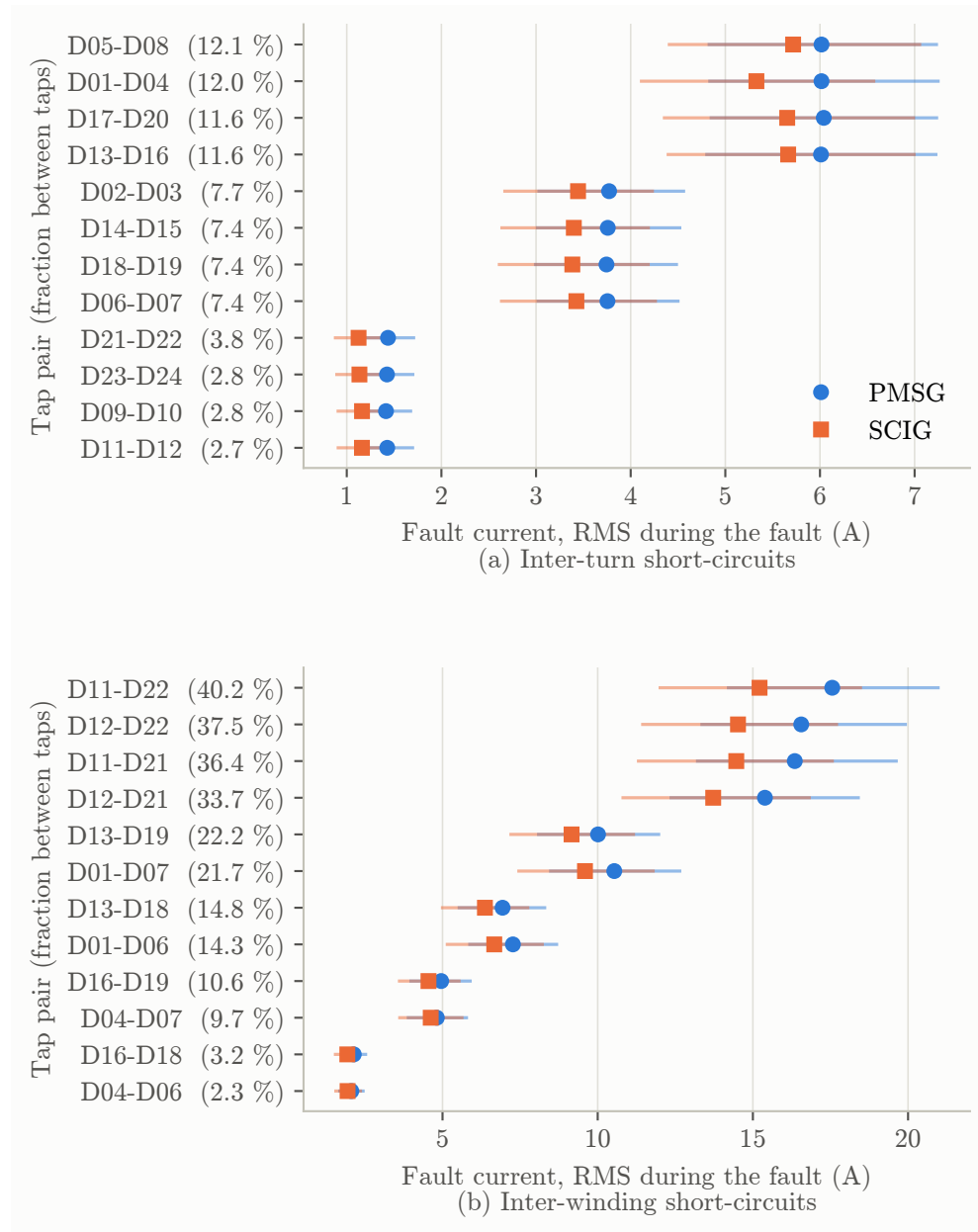


Fig. S1 Fault current per tap pair, PMSG (rated 6.3 A) and SCIG (7.7 A), inter-turn (a) and inter-winding (b) faults: mean over the nine operating points (marker) and range (bar)

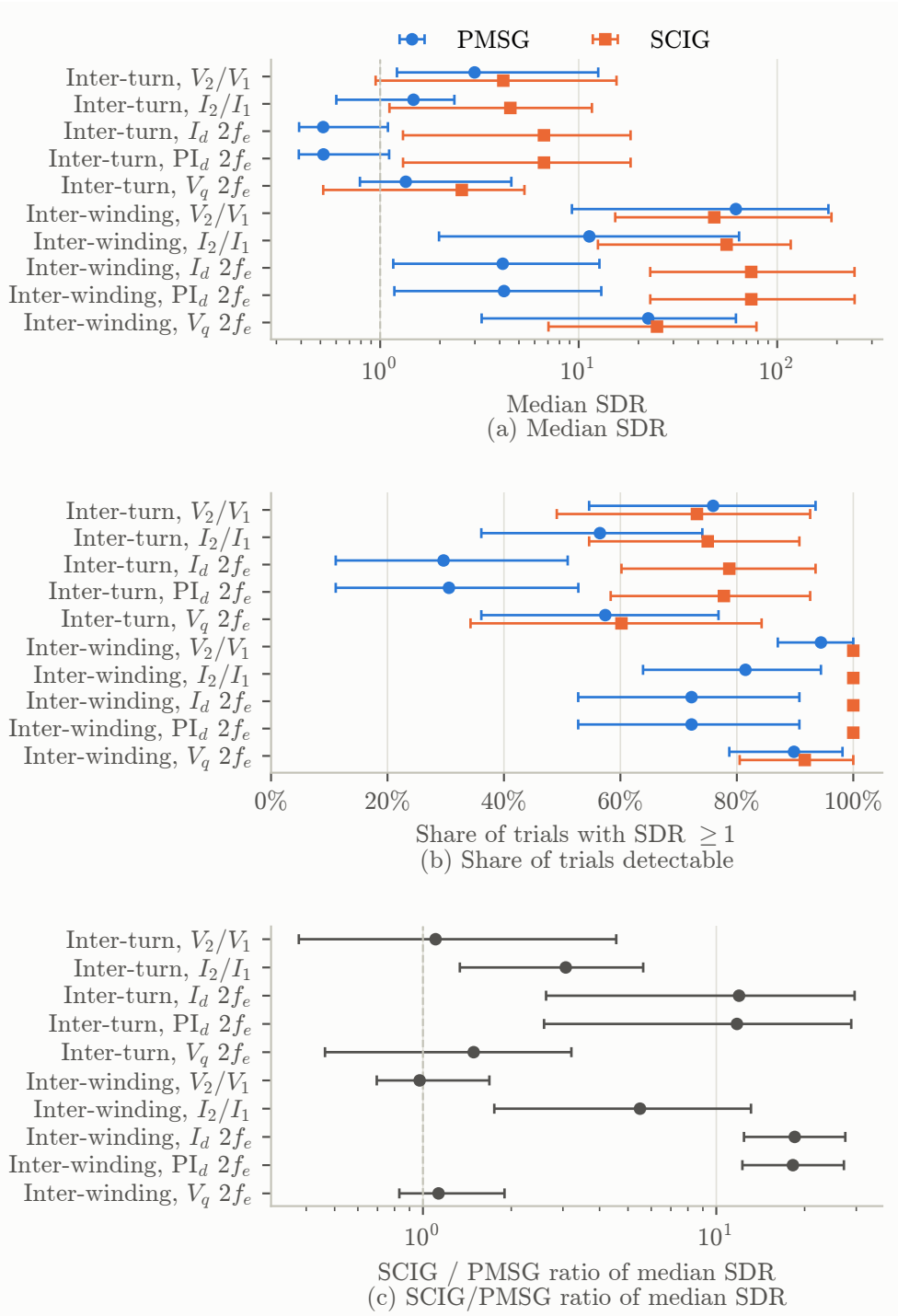


Fig. S2 Tap-pair bootstrap 95% intervals of the median SDR (a), the detectable share (b) and the SCIG/PMSG ratio of median SDR (c), five-cycle windows